

Chapter 160 — Open Radical Prostatectomy

- **Most radical prostatectomies (RPs) in the United States and Europe are now robot-assisted**, although there is no evidence that the robotic approach gives superior cancer control, fewer technical complications, or better urinary or sexual function.
- **The open approach remains relevant**: its lessons transfer directly to robotic surgery, it is needed where no robot is available, and it is the fallback when conversion to open surgery is required.
- **Total surgical removal is the best cure for clinically localised prostate cancer**. RP is the only treatment shown in a randomised controlled trial (Bill-Axelsson; Scandinavian Prostate Cancer Group-4, SPCG-4) to reduce progression to metastases and death from prostate cancer.
- **Better understanding of periprostatic anatomy** has led to less bleeding, fewer technical complications, and higher rates of continence and potency.
- **The goals of surgery, in order of importance**: control the cancer → minimise technical complications → preserve urinary control → preserve sexual function.
- **Open radical retropubic prostatectomy (ORRP) is one of the most technically demanding operations in urology**. Patient selection and operative technique both call for considerable skill and experience.

Concept — *The chapter's order of priorities (cancer, then complications, then continence, then potency) settles every intraoperative trade-off that follows. When nerve sparing and a clear margin conflict, the margin wins.*

RADICAL RETROPUBIC PROSTATECTOMY: SURGICAL ANATOMY

Venous and Arterial Anatomy

Venous drainage

- **Prostatic veins drain into the Santorini plexus**. You need a precise knowledge of it to get a bloodless field at the membranous urethra and prostatic apex.
- **Deep dorsal vein of the penis**: leaves the penis beneath Buck's fascia, between the corpora cavernosa → pierces the urogenital diaphragm → divides into three branches: the **superficial branch** and the **right and left lateral venous plexuses**.
- **Superficial branch**: runs between the puboprostatic ligaments as the central vein over the bladder neck and prostate. It is seen early in retropubic surgery, communicates over the bladder and into the endopelvic fascia, and lies **outside the anterior prostatic fascia**.
- **Common trunk and lateral plexuses are concealed by the prostatic and endopelvic fascia**. They run posterolaterally and communicate freely with the pudendal, obturator and vesical plexuses. Near the puboprostatic ligaments, small branches pierce the pelvic sidewall musculature to join the internal pudendal vein.
- **The lateral plexus contributes to the inferior vesical vein** → internal iliac vein. Because these veins anastomose freely, **any laceration of these friable veins can cause major blood loss**.
- **The common trunk lies directly over the urethra**, and this is where the dorsal vein is transected.

Arterial supply

- **The prostate is supplied by the inferior vesical artery**. After giving small branches to the seminal vesicle and to the base of the bladder and prostate, it ends in two groups:

TERMINAL PROSTATIC ARTERIES: URETHRAL VS CAPSULAR GROUPS

Feature	Urethral group	Capsular group
Entry or course	Enters at the posterolateral vesicoprostatic junction	Runs along the pelvic sidewall in the lateral pelvic fascia, posterolateral to the prostate
Territory	Vesical neck and periurethral gland	Outer prostate (ventral and dorsal branches); ends as a small cluster supplying the pelvic floor
Surgical significance	Divided at the bladder neck (5- and 7-o'clock pedicles)	Surrounded by an extensive nerve network → macroscopic landmark for the microscopic cavernous nerves (the NVB)

- **Main arterial supply to the corpora cavernosa: the internal pudendal artery. Accessory (aberrant) pudendal arteries** can arise from the obturator, inferior vesical or superior vesical arteries. They run along the lower bladder and the anterolateral prostate, so they are divided at RP. This may compromise penile inflow, **especially in older men with borderline penile blood flow.**

Pelvic Plexus

- **The pelvic plexus supplies autonomic innervation to the pelvic organs and external genitalia.** It combines **parasympathetic** preganglionic visceral efferent fibres from the **sacral centre (S2–S4)** with **sympathetic** fibres from the **thoracolumbar centre**, which arrive via the hypogastric nerve.
- **Location:** retroperitoneal, beside the rectum, **5–11 cm from the anal verge**. It is a fenestrated rectangular plate in the sagittal plane, with its **midpoint at the tip of the seminal vesicle**.
- **The inferior vesical vessels perforate the plexus.** So ligating the "lateral pedicle" in its midportion divides both the vessels and the nerves to the prostate, urethra and corpora cavernosa.
- **Visceral branches** go to the bladder, ureter, seminal vesicles, prostate, rectum, membranous urethra and corpora cavernosa. **Somatic motor axons** also pass through the plexus to the levator ani, coccygeus and striated urethral muscle.
- **Prostatic nerves run outside the capsule and Denonvilliers fascia** until they pierce the capsule. **Branches to the membranous urethra and corpora** also stay outside the capsule, in the lateral pelvic fascia dorsolateral to the prostate, between the prostate and the rectum.
- **Neurovascular bundle (NVB), the "NVB of Walsh":** the capsular vessels plus the cavernous nerves. The nerves are microscopic, but the vessels show where they are.
- **The cavernous branches join the capsular vessels in a spray-like distribution, forming the NVB 20–30 mm distal to the prostatovesical junction.** The NVBs lie in the lateral pelvic fascia, **between the prostatic fascia and the levator fascia.**
- **At the apex, the branches to the cavernous bodies and striated sphincter spray both anteriorly and posteriorly,** with wide individual variation. Small nerve branches travel anteriorly, away from the vessels. This is the anatomical basis for high anterior release.
- After piercing the urogenital diaphragm, the nerves pass **behind the dorsal penile artery and nerve** before entering the corpora.

Striated Urethral Sphincter

- **The sphincter is not a horizontal "sandwich".** It is a **vertically oriented tubular sheath** (Oelrich) around the membranous urethra.
- **Development:** in utero it runs continuously from the bladder to the perineal membrane. As the prostate grows from the urethra, it invades and thins the sphincter, and part of the muscle atrophies.
- **Adult sphincter:** at the apex the fibres are **horseshoe-shaped**, forming a tube that fuses in the midline posteriorly near the apex. The prostate does not sit on a flat urogenital diaphragm "like an apple on a

shelf". **The sphincter has broad attachments over the apical prostatic fascia.** This shapes the apical dissection and the urethral reconstruction.

- At the apex, the **NVBs lie posterolateral to the circumferential striated sphincter.**

PASSIVE VS ACTIVE CONTINENCE

Feature	Passive continence	Active continence
Muscle	Striated urethral sphincter	Levator ani (levator urethrae, pubourethralis), which surrounds the apex and membranous urethra and inserts into the perineal body
Fibre type	Fatigue-resistant, slow-twitch	Voluntary contraction
Innervation	Mainly the pudendal nerve for both , with additional somatic motor fibres through the pelvic plexus	

- **Pelvic floor exercises after surgery contract the levator ani.** The striated sphincter shares its innervation, so it is exercised as well.

Pelvic Fascia

THE THREE FASCIAL INVESTMENTS OF THE PROSTATE

Layer	Position	Key features	Surgical rule
Denonvilliers fascia	Between the anterior rectal wall and the prostate; extends cranially over the posterior seminal vesicles	Thickest at the base and seminal vesicles; thins caudally and ends at the striated sphincter. Anterior and posterior layers cannot be told apart microscopically	Excise it completely to get an adequate margin
Prostatic fascia ("capsule")	Anteriorly and anterolaterally, continuous with the parenchyma	Carries the dorsal vein tributaries and the Santorini plexus anteriorly	Must stay on the prostate in nerve sparing
Levator fascia	Covers the pelvic musculature; fuses laterally with the prostatic fascia to form the lateral pelvic fascia ; separates posterolaterally to follow the perirectal muscles	The blood supply and autonomic nerves reach the prostate between the levator and prostatic fasciae	Incised or released to reach the NVB

- **The two approaches meet the fascia differently:**
 - **Perineal:** the lateral and anterior pelvic fasciae are reflected off the prostate, which spares the dorsal vein and Santorini plexus → **less blood loss.**
 - **Retropubic:** the prostate is approached from outside these fasciae → the **dorsal vein complex (DVC) must be ligated and the lateral pelvic fascia divided.**

Concept — *The anatomy comes down to one sandwich. The nerves, arteries and veins all run in the plane between the levator fascia and the prostatic fascia. Leaving the prostatic fascia on the gland and Denonvilliers fascia on its back is what reconciles nerve sparing with clear margins.*

SURGICAL TECHNIQUE

Preoperative Preparation

- **Assessment:** focused review of systems; full medical, surgical and anaesthetic history; all prescription and non-prescription medicines; examination; routine laboratory tests.
- **Rare but life-threatening risks:** myocardial infarction, stroke, arrhythmia, pulmonary embolism, haemorrhage and anaesthetic reactions. Identify at-risk men so that the risk can be reduced or an alternative treatment offered.

- **Factors that make the operation harder but are not contraindications:** previous abdominal or pelvic surgery or irradiation, previous transurethral surgery, extensive biopsies, significant inflammatory bowel disease, previous inguinal or incisional hernia mesh, and prostate size. They call for more skill and experience.
- **Inguinal hernia:**
 - **About 15% of men have a coexisting inguinal hernia** if examined properly before surgery, and **up to 20% develop one after ORRP**. Most were asymptomatic beforehand, so ORRP appears to turn asymptomatic hernias into symptomatic ones.
 - → **Examine standing, with a Valsalva manoeuvre** → repair any hernia found by **preperitoneal repair at the time of ORRP**. Standing Valsalva ultrasonography picks up even more subclinical hernias.
 - With thorough evaluation and preperitoneal repair, symptomatic hernias after ORRP are rare.
- **Timing: defer surgery 6 weeks after needle biopsy and 12 weeks after transurethral resection of the prostate (TURP).** This lets inflammatory adhesions and haematoma settle so that the tissue planes are near normal, which matters most for NVB preservation and avoiding rectal injury.
- **No autologous blood donation.** Transfusion rates are very low, and donated units go unused unless the patient is anaemic, so donation only worsens postoperative anaemia. **Preoperative erythropoiesis-stimulating agents** raise the haematocrit and reduce anaemia after prostatectomy. A higher haematocrit at discharge speeds recovery.
- **Antibiotic prophylaxis for open or laparoscopic surgery entering the urinary tract:** a **first- or second-generation cephalosporin**, or an **aminoglycoside plus metronidazole or clindamycin**.
- **Anticoagulants and antiplatelet agents:** plan with the internist or cardiologist. **Drug-eluting coronary stents** set the risk of **stent thrombosis, which is usually catastrophic**, against the risk of perioperative bleeding. **Low-dose aspirin** often gives adequate short-term antiplatelet cover without increasing bleeding at ORRP.
- **Bowel preparation:** clear fluids the day before → half a bottle of magnesium citrate that evening → enema on the morning of surgery. Admission on the day of surgery.
- **Pretreatment magnetic resonance imaging (MRI)** can guide how much of the NVB and membranous urethra to preserve, to improve potency and continence.

Special Instruments

- **Fibreoptic headlight: essential**, because much of the operation takes place beneath the pubis where the view is poor.
- **Balfour retractor** with narrow and wide malleable blades. It helps during lymphadenectomy and is **required during RP** for cranial and posterior retraction of the peritoneum and bladder.
- At the surgeon's discretion: coagulating forceps, small, fine and standard right-angled clamps, Metzenbaum and Jamison scissors, and **2.5–4.5× loupes**, which help with the apical dissection.

Anaesthesia, Incision, and Lymphadenectomy

- **General endotracheal anaesthesia is preferred.** Keep **relative hypotension (systolic ≤ 100 mm Hg)** and **limit crystalloid to 1500 mL until the prostate is out**.
- **Position:** supine. Flex the table in obese men to lengthen the umbilicus–pubis distance.
- **Catheter: 16-Fr silicone (Silastic) Foley catheter**, balloon 20 mL, on closed drainage. The smaller calibre makes it easier to place urethral mucosal sutures. A right-handed surgeon stands on the patient's left.
- **Incision sequence:** extraperitoneal lower midline incision from the pubis towards the umbilicus → anterior fascia opened to the pubis → recti separated → transversalis fascia opened sharply → **space of**

Retzius → peritoneum mobilised off the external iliac vessels to the common iliac bifurcation, **without dividing the vas deferens** → Balfour retractor placed.

- **Preserve the soft tissue over the external iliac artery.** It carries the lymphatics from the lower limb, and dividing them risks **leg oedema and lymphocele**.
- **Exposure for lymphadenectomy:** narrow malleable blade under the mobilised vas to push the peritoneum up; deep Deaver retractor to pull the bladder medially.
- **Pelvic lymph node dissection (PLND) before RP for intermediate- and high-risk disease.** It is omitted for well-differentiated or low-volume favourable intermediate-risk disease (Gleason Grade Group 1–2).
- **PLND sequence** (start on the side of the dominant tumour): divide the adventitia over the external iliac vein, **sparing the lymphatics over the external iliac artery** → beneath the vein to the pelvic sidewall → down to the femoral canal, ligating the lymphatics (**the node of Cloquet need not be removed**) → up the sidewall to the common iliac bifurcation, clearing the angle between the external iliac and hypogastric arteries → **obturator nodes**, protecting the **obturator nerve**; obturator vessels skeletonised but not ligated unless bleeding → down to the pelvic floor, exposing the hypogastric veins → repeat on the other side.
- **This is an extended dissection.** It removes more nodes than a limited one, **improving staging with a possible therapeutic benefit** in some men.

Exposure

- **Displace the peritoneum superiorly with a malleable blade**, which also pushes the bladder gently posteriorly, to expose the anterior prostate.

Incision in the Endopelvic Fascia

- Clear the fibroadipose tissue → expose the pelvic fascia, the puboprostatic ligaments and the superficial dorsal vein.
- **Open the endopelvic fascia where it reflects onto the pelvic sidewall, well away from the bladder and prostate**, at the point where it is transparent over the levator ani.
- **Incising too medially enters the lateral Santorini plexus → persistent venous bleeding.** The prostatic arteries and pelvic plexus branches lie beneath this plexus.
- Extend the incision anteromedially towards the puboprostatic ligaments → the lateral prostate can now be palpated.
- **Small pudendal perforating vessels** through the pelvic musculature → **clip, do not coagulate**. The pudendal artery and nerve lie just deep to the muscle along the pubic ramus.

Division of the Puboprostatic Ligaments

- Tease the fibrofatty tissue off the superficial dorsal vein and ligaments → dissect the superficial vein from the medial ligament edge → coagulate and divide it → push the prostate posteriorly with a sponge stick → divide each ligament with scissors.
- **Go deep enough to expose the junction of the prostatic apex with the anterior surface of the DVC**, where the complex will be divided.
- **Spare the pubourethral component**, which keeps the striated sphincter fixed anteriorly to the pubis.

Preservation of Accessory Pudendal Arteries

- **Arterial insufficiency contributes to erectile dysfunction after nerve-sparing RP.** Accessory pudendal arteries over the anterolateral prostate are one source of it.

- **Prevalence varies by method:** common in cadaveric dissection (about 70%), less common on selective internal pudendal angiography (about 7%), and **large visible vessels in about 4% of men.**
- **Releasing them can cause brisk bleeding** that forces prompt DVC ligation. So **complete the contralateral side first:** if a prominent vessel is unilateral, divide the endopelvic fascia and puboprostatic ligament on the opposite side first.
- **Sequence:** divide the endopelvic fascia lateral to the vessel and the puboprostatic ligament (the vessel runs beneath it) → elevate the artery on a vessel loop → sharp and right-angled-clamp release from its investing fascia, elevating the anterior prostatic fascia → divide the venous tributaries (venous bleeding is controlled with a running 4-0 absorbable suture) → carry the dissection caudally beyond the planned DVC division point.

Ligation and Division of the Dorsal Vein Complex

- **Aim:** divide the complex with minimal blood loss, **without injuring the striated sphincter or entering the anterior apex.**

CONTROL OF THE DORSAL VEIN COMPLEX: STAPLER VS SUTURE LIGATION

Feature	Endovascular stapler (current preference)	Suture ligation (traditional)
Sequence	Babcock bunches the endopelvic fascial edges → Endo GIA (vascular load) advanced about 4 cm over the DVC → move the Foley in and out: fire only if it moves freely (urethra not caught) → deploy → residual vein oversewn with 3-0 Monocryl	Sponge stick pushes the prostate back → 3-0 Monocryl through the DVC just distal to the apex → needle reversed → through the perichondrium of the pubic symphysis → figure-of-eight horizontal mattress tied; suture kept long → Babcock bunches the fascia → figure-of-eight 2-0 suture on the anterior prostate near the bladder neck (proximal back-bleeding, worse with incompetent venous valves)
Blood loss	About 200–400 mL less than traditional ligation	Controls much of the venous bleeding
Other advantages	Less traction → less stretch injury to the NVBs and sphincter; three staple rows each side haemostase both ends; apical staple line gives extra cover over anterior apical tumours	Flat, unbunched surface is easier to divide; recreates puboprostatic support for the sphincter; fixes the DVC anteriorly , showing the anterior apical plane

- **Surgeon's stance for all apical sutures:** face the head of the table, needle driver against the pubis, perpendicular to the patient.

Division of the Dorsal Vein Complex

- Downward sponge-stick pressure on the anterior prostate → divide with Metzenbaum scissors or a No. 15 blade, **starting at the left edge**, where the apical junction is best seen. Because the distal complex is fixed anteriorly, the plane between the anterior prostate and the striated muscle opens up. Loupes help to tell the tissues apart, and resting the scissors on the anterior apex helps to find the plane.
- **The anterior apex is the commonest site of positive surgical margins**, because its surface is hard to identify.
- **Caveat:** near the apex the NVB is often **tethered medially beneath the sphincter by an apical vessel**. Dividing the sphincter too close to the apex risks the NVB. → **Divide the lateral sphincter edges obliquely, midway between the apex and the pelvic floor.**
- **Venous control must be absolute** so that the rest of the operation is bloodless. Use the kept 3-0 Monocryl (5/8-circle needle) to **oversew the distal sphincter–DVC edge with a loosely held driver, forming a hood over the anterior urethra**. Circumflex veins at 5 and 7 o'clock → hemoclips.

- **Proximal back-bleeding:** oversee the proximal DVC edges on the anterior prostate in a V with the preplaced 2-0 suture. This spreads tension evenly across the prostatic fascia, which helps a high release.

Division of the Urethra and Placement of Urethral Sutures

- **Timing:** for **standard nerve release, divide the urethra first**. For more aggressive nerve preservation (high anterior release), divide the urethra **after** nerve release.
- **Sequence:** sponge stick pushes the prostate back → release the lateral bands of the striated muscle at their midpoint → **right-angled clamp around the urethral smooth muscle close to the apex**, to transect as near the apex as possible → divide the **anterior two-thirds** of the urethra, sparing the catheter → place sutures → remove the catheter and place the 6-o'clock suture → divide the posterior urethral band.
- **The clamp manoeuvre separates the posterior urethra from the posterior striated sphincter**, and in a dry field it shows how the posterior sphincter relates to the NVB. The intact **posterior sphincter protects the NVBs**.
- **Six 3-0 Monocryl sutures** (5/8-circle tapered needle) in the distal urethra. The text gives 1, 3, 5, 7, 9 and 11 o'clock; the operative figures show 12, 2, 5, 6, 7 and 10 o'clock.
- **Mucosa and submucosa only, not smooth muscle. Smooth-muscle bites delay the return of continence.** The 16-Fr catheter makes the mucosa easy to see.
- If the urethra is flimsy → include the DVC in the first (anterior) suture for tensile strength. The first sutures lift the mucosa and make the rest easier.
- Sutures go outside → in. If inside → out is easier, a **French-eye needle** passes them through the lumen from the bladder side at the end. Sutures are covered with towels to prevent traction.

Posterior striated sphincter complex (*skeletal muscle plus fibrous tissue*). *Precise division matters for four reasons:*

- (1) **adequate apical margins**; (2) finding the **correct plane on the anterior rectum so that all of Denonvilliers fascia is excised**; (3) **avoiding blunt trauma to the NVBs** immediately behind it; (4) **preserving continence**.
- **Sequence:** right-angled clamp beneath the **left** edge, **midway between the apex and the urethra**, where the NVB has fallen posteriorly beneath the clamp → divide → repeat on the **right** → divide the **central** part last.
- **Divide from each side, never all from one side.** A single pass from one side risks the contralateral NVB. A clamp passed too close to the apex risks the NVB.

Identification and Preservation of the Nerve Bundle

- In most surgical candidates both NVBs can be preserved safely; excising both is rarely necessary. With modern technique and phosphodiesterase type 5 (PDE5) inhibitors, most healthy potent men **under 65** should remain potent.
- **Extracapsular extension rarely goes deeper than 1–2 mm**, and that much tissue is often present even when the NVB is kept.
- **The nerves directly on the capsule enter the prostate; they are not the cavernous nerves.** The cavernous branches lie **posterior to the capsular vessels**.

NVB DISSECTION PLANES COMPARED

Plane	Where the dissection runs	Margin risk	Verdict
Interfascial	Between the levator fascia and the prostatic fascia; prostatic fascia left on the prostate	Low	Correct nerve-sparing plane

Plane	Where the dissection runs	Margin risk	Verdict
Intrafascial	Beneath the prostatic fascia, directly on the parenchyma , to avoid capsular-vessel bleeding	High	Never used
Wide excision	NVB and lateral pelvic fascia taken with the specimen	Does not lower the overall margin rate	For palpable induration, a fixed bundle or an uncertain posterolateral margin

- **Excising the NVB does not prevent positive margins**, because the NVB is not where most margins occur. **Positive margin sites, commonest first: apex → posterior → posterolateral**. About a fifth of positive margins are multiple. **Nerve-sparing and non-nerve-sparing procedures have the same positive margin rate**.
- **Walsh series**: NVBs preserved bilaterally in 87% and one NVB excised in 13%, with an overall **positive margin rate of about 5%**. Excision of both NVBs is almost never needed. **Capsular penetration bad enough to need both excised implies probable distant spread and incurability**, and in practice one side usually turns out to be free.
- **The decision is never final before surgery**. Inputs: **erectile status** (continence is the same after bilateral and unilateral nerve sparing, and whether or not erections return, so the NVBs are preserved less aggressively in men who already have erectile dysfunction); **palpable apical lesion**; **high predicted capsular penetration on Partin tables** or preoperative MRI.
- **Intraoperative triggers for wide excision on that side**: **palpable induration in the lateral pelvic fascia** when the endopelvic fascia is opened; **too little tissue over the posterolateral surface after removal** (the NVB can then be excised secondarily).

Identification of the Neurovascular Bundle

- **Tactile feedback is the advantage of open surgery**. **Palpable induration → excise the bundle; a bundle fixed to the prostate on gentle release → do not preserve it**. Tactile feedback is dulled in laparoscopic surgery and absent in robot-assisted surgery.
- **No upward traction** on the prostate during release. Roll it from side to side instead. With the catheter out, the prostate is softer and the release plane is easier to find.

Standard Preservation of the Neurovascular Bundles

- **Starts after** urethral division, urethral suture placement and division of the posterior sphincter.
- **Sequence**:
 - Babcock or sponge stick turns the prostate on its side, so the **prostate is released from the bundle, not the bundle from the prostate**.
 - Right-angled clamp under the **superficial levator fascia, starting at the bladder neck** (where a thick band, once divided, frees the prostate at once) → on to the apex. **Loupes** confirm the prostatic fascia is intact and allow a gentler dissection.
 - Find the NVB by a **subtle "groove" on the posterolateral prostate** → trace it to the apex, where the NVB turns caudolaterally towards the urethra.
 - Find the medial border at the apex → develop the midline plane posteriorly to the rectum.
 - Release the bundle **from the rectal surface, apex → base**, spreading a right-angled clamp → clip and divide the prominent apical vessels → continue to the midpoint, then to the seminal vesicle.
 - Find the **posterior arterial branch running from the NVB over the seminal vesicle to the prostatic base** → ligate and divide on each side → the NVBs fall away posteriorly, untethered.
- **Apical kinking artefact**: with the prostate rolled across the midline, the NVB looks kinked at the apex and **can be transected by mistake**. It straightens once the apical vessels are divided.

- **Vessel control on the NVB: small hemoclips placed parallel to the bundle. Never any thermal energy** (monopolar, bipolar or harmonic scissors) on the NVB or its branches. The prostatic side of tiny vessels is usually divided with fine scissors, unclipped.
- **Fixation not explained by vascular branches → excise the NVB.**
- **Reasons to start at the apex:** vessel branches are seen before they are cut; an **anteriorly placed bundle** could otherwise have its groove mistaken for the prostatorectal space; lateral veins joining the Santorini plexus to the NVB are easier to manage.
- Dissecting from the rectal side **keeps Denonvilliers and prostatic fascia on the gland**; only fragments of levator fascia come away laterally. Margins caused by cutting into organ-confined tumour then become rare, and **partial bundle excision with a negative margin and preserved potency is possible even with extraprostatic extension near the NVB.**

High Anterior Release of the Neurovascular Bundles at the Apex

- **Purpose:** faster return of potency and continence, by **reducing traction** on the nerves to the cavernous bodies and sphincter and avoiding **transection of small anteriorly running branches.**
- **Risk: less soft tissue at the apex**, so the positive margin rate may rise. Candidates must therefore have a low risk of extraprostatic extension near the NVB.

TSUZUKI CRITERIA: EXTRAPROSTATIC EXTENSION NEAR THE NVB (PER SIDE)

Criterion	Threshold
Prostate-specific antigen (PSA)	>10 ng/mL
Gleason score	>6
Average percentage of core involved, that side	>20%
Percentage of cores with tumour, that side	>33%
Perineural invasion	Present
Digital rectal examination	Abnormal

- **Two or more criteria → >10% risk of extraprostatic extension near the NVB on that side → not a candidate for high release on that side.**
- **Sequence:** after DVC ligation, before urethral incision, and only with complete haemostasis → incise the **levator fascia over the anterior apex** along the lateral edge of the DVC → extend distally along the lateral apex, **preserving the glistening white prostatic fascia** beneath the lateral veins → control venous tributaries with **hemoclips, never cautery** → release the levator fascia from the lateral prostatic "shoulders" → beyond the apex the NVB is exposed lateral to the urethra → release it from the lateral prostate to just above the seminal vesicles → divide the DVC to the urethra between the two preplaced sutures.
- **Then:** transect the urethra and place sutures → divide the posterior sphincter → if the release was unilateral, pass a right-angled clamp under the levator fascia at the contralateral apex to enter the same plane from inside.
- Excellent visualisation and magnification are essential to avoid entering the prostate.
- **Unilateral high release recovers potency at 12 months as well as bilateral release → now done routinely on one side only, the side with the more favourable pathology.** This implies the benefit comes from **more accurate, lower-traction preservation**, not from sparing anterior nerve branches.

Wide Excision of the Neurovascular Bundles

- **Free the contralateral NVB from the apex first**, to avoid traction injury during the wide excision.

- **Sequence:** identify the NVB at the apex → right-angled clamp passed **medial → lateral, directly on the anterior rectal surface** → divide the bundle **without ligation**, to take maximum soft tissue (clip the distal end later if it bleeds) → divide the fascia on the lateral rectal surface from apex to base → end at the **tip of the seminal vesicle, where the NVB is ligated and divided**.
- **Lateral → medial passage of the clamp makes rectal injury more likely.**
- The NVB and lateral pelvic fascia are excised under direct vision, giving generous soft-tissue cover over the primary tumour.

Posterior Dissection and Division of the Lateral Pedicles

- Replace the catheter → **light upward traction on the catheter** → divide the attachment between the rectum and Denonvilliers fascia in the midline, **leaving all layers of Denonvilliers fascia over the seminal vesicles**.
- **Never place the sponge stick on the prostate**, because displaced tissue can produce a **false-positive margin**. Put it on the catheter.
- **Divide the lateral pedicle on the lateral surface of the seminal vesicles** (the NVB has already fallen away) in **three sequential layers: superficial → middle → deep (beside the seminal vesicle)**. **Dividing it all at once risks cutting into the prostate**.
- Arterial bleeders → hemoclips. More soft tissue stays on the prostate, and the NVBs are protected.
- Continue onto the anterolateral bladder–prostate junction → divide Denonvilliers fascia over the seminal vesicle tips. Many surgeons divide the vasa and free the seminal vesicles at this stage.

Division of the Bladder Neck and Excision of the Seminal Vesicles

- **Sequence:** incise the **anterior bladder neck at the prostatovesical junction** → down to and through the mucosa → deflate the Foley balloon and clamp the two catheter ends together for traction → widen the incision → divide the **inferior vesical pedicle branches at 5 and 7 o'clock** → find the plane between the **anterior seminal vesicles and the posterior bladder wall** → divide the posterior bladder neck with scissors hugging the seminal vesicles, **watching the ureteric orifices** → Allis clamp on the bladder neck → clip and divide the vasa → free the seminal vesicles → ligate the arteries at each seminal vesicle tip → divide the remaining Denonvilliers attachments → remove the specimen.
- **The pelvic plexus lies on the lateral surface of the seminal vesicles.** Dissect under direct vision, especially laterally, **staying close to the seminal vesicle** and clipping small arteries.
- **Complete excision of the seminal vesicle is essential for cancer control.**
- **Inspect the specimen: any doubt about the posterolateral margin → excise the NVB on that side.**
- **Haemostasis near the NVB: hemoclips, no cautery.** A rectovesical haematoma can cause inflammation around the NVBs that **delays the return of sexual function**. → A **small peritoneal window in the rectovesical cul-de-sac** lets any haematoma decompress.

Bladder Neck Reconstruction and Anastomosis

- **Tennis-racquet closure:** running or interrupted **2-0 absorbable, full-thickness** muscularis plus mucosa (including the mucosa prevents troublesome haematuria), **from the posterior midline forwards**, until the neck matches the urethral diameter.
- **Ureteric orifices:** previously injected indigo carmine helps show them. Ureteric catheters are usually unnecessary.
- **Mucosal eversion:** interrupted or running **4-0 absorbable** sutures advance the mucosa over the raw muscle → a rosette of mucosa → **mucosa-to-mucosa anastomosis**. The **6-o'clock suture is left long** to aid urethral suture placement.

- **Bladder neck intussusception** (buttressing sutures stop the neck pulling open as the bladder fills):
 - **Sequence:** remove any hemoclips in the area (they can fold into the neck and cause contracture) → **2-0 Maxon in the posterior bladder wall edges about 2 cm from the reconstructed neck**, tied in the midline → slacken the malleable blade to show loose perivesical tissue → **second figure-of-eight 2-0 Maxon anteriorly, about 2 cm lateral to the neck on each side**, tied.
 - **End point:** "turtle sign", with the bladder neck poking out beneath the anterior hood like a turtle's head from its shell. Fill with saline → should be competent with little leak → empty completely with forceps through the neck.
 - With intussusception, most men are pad-free by 3 months and nearly all by 1 year.
- **Margin–continence trade-off:** this technique favours **ample apical margins** and may remove more striated muscle, so some men take longer to become pad-free. Some surgeons preserve the puboprostatic ligaments and dissect closer to the apex, which may spare sphincter but risks **inadequate apical margins**.
- **Anastomosis sequence:** check haemostasis → **new 16-Fr silicone Foley catheter** → pass each of the six 3-0 Monocryl sutures through the bladder neck at its matching position, **inside → out** (traction on the 6-o'clock suture everts the mucosa; French-eye needle for the 12-, 2- and 5-o'clock sutures) → irrigate the catheter and test the balloon → pass the catheter and inflate with 15 mL → **Babcock clamp on the anterior bladder wall holds the neck down in the pelvis while tying (no catheter traction)** → **tie the anterior suture first**, then 2, 5, 10, 7 and 6 o'clock.
- **No tension on the anastomosis.** If there is tension → mobilise the bladder off the peritoneum. Babcock coaptation has **markedly reduced bladder neck contracture**.
- **Closure:** check the catheter is not caught → irrigate → small suction drain through the fascia away from the midline, running **between (not through) the recti** → running No. 2 nylon to the fascia, skin clips → catheter taped to the thigh.

Postoperative Management

- **Recovery is usually uneventful.** Outcomes improved with the Walsh technique: **mean stay about 1.7 days**; urine leak, transfusion, ileus and re-exploration for haemorrhage are all well under 1%.
- **Early mobilisation** the evening of surgery; discharge on day 1–2. Clear fluids that evening, low-fat diet the next day. **Single closed-suction drain** removed at discharge or when output falls below 50 mL/day off suction.
- **Analgesia:** intravenous patient-controlled opioid (hydromorphone, fentanyl or morphine) or ketorolac. **Ketorolac is preferred**, to reduce opioid use. Intravenous paracetamol works similarly but is costly. The combination reduces ileus.
- **Criteria for intravenous ketorolac** (non-steroidal anti-inflammatory drug, NSAID): **creatinine <1.2 mg/dL, estimated blood loss <1500 mL, no uncontrolled diabetes, no postoperative haematuria**. Given before transfer to recovery or the next morning.
- **Catheter:** very bothersome and delays return to activity and work → remove as soon as safely possible (the risks are retention and bladder neck contracture). **Never remove it with significant extravasation.**
- **Most anastomoses show no extravasation on cystography by day 4**, but removal then causes **frequent acute retention**, so it is not recommended.
- **Delayed anastomotic healing is almost always caused by a perianastomotic haematoma distracting the anastomosis.**
- **Catheter removal options at 1 week:**
 - **Routine cystography at 1 week** allows early removal and identifies the few men who need longer drainage.

- **Transperineal saline sonocystography** avoids contrast cost and radiation.
- **Remove at 7 days without imaging if the haematocrit falls by <6% between recovery and discharge.**
- **Bladder spasms** usually settle, and reassurance is enough. If severe: **oral oxybutynin 5 mg** (anticholinergic) or **oral diazepam 5–10 mg**.

Concept — Every step of the technique does one of three jobs: gain absolute venous control before touching the apex; keep the correct fascia on the specimen; avoid traction and heat on the NVB. Apex-first, side-to-side, clip-not-cautery and stapled or sutured DVC control all follow from those three rules.

COMPLICATIONS

- **ORRP is well tolerated**, with minimal morbidity and **low mortality (0.2%)**.

Intraoperative Complications

INTRAOPERATIVE COMPLICATIONS: CAUSE AND MANAGEMENT

Complication	Typical setting	Management
Haemorrhage (commonest; usually venous)	Endopelvic fascia incision, puboprostatic ligament division, apical exposure and DVC transection	Packing, exposure, suture or clip ligation; fine 5-0 cardiovascular sutures if needed. Troublesome DVC bleeding → divide the DVC completely over the urethra and oversee the end (the single best measure)
Obturator nerve injury	PLND	If divided → reanastomose with fine non-absorbable sutures
Rectal injury	Apical dissection of small prostates , developing the rectum–Denonvilliers plane	Primary two-layer repair with omental interposition (below); diverting colostomy after previous radiotherapy (salvage RP)
Ureteric injury	Dissection within the trigone while seeking the bladder–seminal vesicle plane	Ureteric reimplantation

- **Why partial DVC division worsens bleeding:** the traction needed for exposure pulls open partly divided veins. Once the vein is fully divided and suture-ligated, bleeding is usually controlled.
- **Blood loss:** with careful technique, **typically 300–1000 mL**, and **homologous transfusion is rarely needed**. Autologous donation is no longer required, but **every patient has a type and screen**.
- **Rectal injury repair sequence:** complete the prostatectomy → reconstruct the bladder neck → perfect haemostasis → small peritoneal opening → **fashion a slender, well-vascularised omental pedicle** long enough to reach the pelvic floor → pass it through an opening in the rectovesical cul-de-sac → assistant **dilates the anal sphincter digitally** to show the injury → **freshen the edges and close in two layers** → **fix the omentum over the repair, between the rectal closure and the vesicourethral anastomosis** → anastomosis → copious antibiotic irrigation → **several days of broad-spectrum aerobic and anaerobic cover**.
- **Omental interposition prevents rectourethral fistula.** In the reported series, no patient developed a wound infection, pelvic abscess or fistula.

Postoperative Complications

- **Life-threatening delayed haemorrhage is rare. Definition of significant postoperative bleeding:** needing acute transfusion to support blood pressure. Most cases are managed expectantly, and exploration is rarely needed.
- **Explored vs conservative:** similar blood product requirements, but **shorter hospital stay after re-exploration**. With **non-operative management**, the pelvic haematoma can drain through the anastomosis

→ **bladder neck contracture and long-term incontinence**. Only a minority of explored men have prolonged mild incontinence.

- **Signs of major bleeding in recovery** (hypotension, tachycardia, falling urine output, lower abdominal distension) → **early exploration to evacuate the pelvic haematoma**. This reduces prolonged ileus, anastomotic dehiscence, prolonged catheterisation and bladder neck contracture. **Immediate exploration after fluid resuscitation** often still allows discharge on day 1–2.

Thromboembolic Events

- **Deep vein thrombosis (DVT) with pulmonary embolism (PE) is the major potential cause of death after RP.**
- **Highest risk when PLND is performed, and peak risk 14–28 days after surgery → vigilance for up to 4 weeks.**
- **Prevention:** careful positioning to avoid compressing lower-limb veins, **intermittent pneumatic compression, early ambulation**. Mini-dose unfractionated or low-dose low-molecular-weight heparin (LMWH) is used in some centres but not by the authors, whose DVT and PE rates are very low.
- **An informed patient is the best protection.** Give repeated verbal and written warnings to call at once with: calf or leg swelling or pain; pleuritic chest pain; haemoptysis; breathlessness; sudden weakness or syncope.
- **Treatment: twice-daily subcutaneous LMWH without monitoring is at least as effective and safe as adjusted-dose intravenous unfractionated heparin.**

Bladder Neck Contracture

- **Historical incidence 0.5–10%, falling sharply over the last decade.**
- **Causes:** inadequate mucosal approximation, urinary extravasation, **distraction of the bladder neck by haematoma**.
- **Typically presents at about 3 months, with a poor stream or regression of continence.**
- **Treatment ladder:** cystoscopic dilatation → **cold-knife incision at 3, 6 and 9 o'clock** + short-term intermittent self-catheterisation → recalcitrant cases: **triamcinolone acetonide (200 mg in 5 mL) injected after incision** → very rarely, major reconstruction with an **artificial urinary sphincter**.

Urinary Incontinence

- **Mechanism: usually intrinsic sphincter deficiency.** Few men over 65 (population registry data) need an incontinence procedure within 2 years.
- **Predominant cause: sphincter injury during DVC ligation and division.** Other factors: a poorly developed sphincter, or in older men a thinner, more collagenous one; smooth muscle damaged by large deep anastomotic sutures or denervated by NVB injury; **bladder neck contracture or an over-wide bladder neck**.
- **Prevention:** preserve the sphincter during the apical dissection; a **tension-free** anastomosis; a **small, supple bladder neck**; a precise **mucosa-to-mucosa** anastomosis.
- **Pre-existing bladder outlet obstruction** (detrusor hypertrophy, poor compliance) → slower recovery. **Avoid excessive intraoperative traction on the bladder.**
- **Conservative advice until dry:** restrict fluids; avoid caffeine and alcohol; **stop α -adrenergic antagonists** taken for hypertension or benign prostatic hyperplasia (BPH). Depending on severity and cause, **imipramine, phenylephrine, anticholinergics or adrenergic agonists** may help.
- **Definition of continence used: 0–1 pad per 24 h.**

RECOVERY AFTER ORRP BY AN EXPERIENCED SURGEON (LEPOR CONTINENCE; WALSH POTENCY)

Time after surgery	Continent (0–1 pad/24 h)	Potent (Walsh series)
3 months	80%	38%
6 months	90%	54%
12 months	93%	73%
18 months	—	86%
24 months	97%	—

- **What the table shows: continence recovers faster and more completely than potency.** Both improve steadily, so patients should be counselled that recovery is gradual.
- **The trap:** these are single-surgeon, centre-of-excellence figures. Another expert centre reported return to baseline potency in **only about a third at 2 years** after bilateral nerve sparing, despite PDE5 inhibitors.

Erectile Dysfunction

- **Four predictors of recovery: age <65, preoperative potency, no adult-onset diabetes mellitus, bilateral NVB preservation.** Age at surgery independently affects recovery.
- **Walsh series** (validated questionnaire, independent third-party review): **86% had unassisted intercourse, with or without sildenafil, at 18 months.** About a third used sildenafil, but only 4% could not have intercourse without it.
- **Most recovery happens within 2 years**, but meaningful improvement continues **up to 4 years.**
- **High anterior release → significantly earlier recovery and return to baseline function.** Most potent men are potent at 12 months, and most return to their baseline function, including men having intercourse more than once a week. The same result with unilateral and bilateral release → the benefit comes from **less traction and more accurate preservation**, not from sparing anterior branches. Other surgeons report similar gains with early nerve release.

TRIALS OF PDE5 INHIBITOR PENILE REHABILITATION AFTER RP

Trial	Design	Arms	Result
Padma-Nathan	First multicentre randomised placebo-controlled trial	Nightly sildenafil 50 mg vs placebo for 9 months	Better erectile function at 1 year, off drug: first evidence for prophylactic PDE5 inhibitor use
REINVENT (Montorsi)	Multi-institutional, randomised, double-blind	Vardenafil on demand vs nightly vs placebo for 9 months	On-demand better than nightly at the end of blinded treatment
Sildenafil schedule trial	Randomised, similar design	On-demand vs nightly sildenafil	No difference; trend favouring on-demand

- **Pattern:** the first trial established that rehabilitation helps. **No later trial showed nightly dosing to be better than on-demand dosing**, so daily dosing is not evidence-based.
- **Under-reported sexual effects: penile shortening and climacturia.**

Lower Urinary Tract Symptoms

- **ORRP improves lower urinary tract symptoms (LUTS)** in men with **moderate-to-severe International Prostate Symptom Score (IPSS)** before surgery.
- These men usually gain urinary quality of life **despite the risk of incontinence.** The **benefit lasts up to 15 years**, which suggests that RP changes the natural history of BPH and LUTS.

Concept — *Most late complications trace back to one event: a pelvic haematoma. It distracts the anastomosis, causes bladder neck contracture and incontinence, and inflames the NVBs. That is why the chapter favours early re-exploration, clips near the bundles and a peritoneal window.*

SUMMARY OF OPEN RADICAL RETROPUBIC PROSTATECTOMY

- **More men are now diagnosed with curable disease at a younger age, and men live longer: a man aged 65–70 has a 50% chance of living another 15 years.**
- **ORRP is ideal for men who can be cured and will live long enough to benefit.** These are also the men with the best quality of life after surgery.
- **Robot-assisted laparoscopic RP has spread over the past decade. Its main advantage is less blood loss,** and even its proponents concede there is no meaningful overall advantage. Some of the authors still operate exclusively by the open approach.

KEY POINTS: ANATOMIC OPEN RADICAL RETROPUBIC PROSTATECTOMY

- There is **no better cure for organ-confined prostate cancer than total surgical removal.**
- **Goals, in order:** cancer control → minimise technical complications → continence → potency.
- **Defer surgery 6 weeks after needle biopsy and 12 weeks after TURP.**
- A **fibreoptic headlight** is the most useful instrument, because so much of the operation takes place beneath the pubis.
- **Well-differentiated or low-volume favourable intermediate-risk disease (Grade Group 1–2) → no PLND.**
- Nerve branches to the sphincter and cavernous bodies may run anteriorly at the apex → **high anterior release before DVC division** → earlier recovery of potency.
- **Both NVBs can usually be preserved safely;** excising both is rarely necessary.
- **Tactile feedback:** palpable induration in the lateral pelvic fascia → excise the bundle; a bundle fixed to the prostate → do not preserve it. Tactile feedback is dulled in laparoscopic surgery and absent in robotic surgery.
- **Acute transfusion for hypotension after RP → explore and evacuate the pelvic haematoma,** to reduce bladder neck contracture and incontinence.
- **Thromboembolism with PE is the major cause of death after RP.** Prevention: careful positioning, intermittent compression devices, early ambulation.
- **Complete excision of the seminal vesicles is essential for cancer control.**
- ORRP is ideal for men who can be cured and will live long enough to benefit.